

End-to-End Verification Test Results Report

E2E Testing & Verification Report

A complete verification test cycle of the **Titanic Survival Intelligence Suite** was executed. The results and coverage are outlined below.

Test Summary

Test Phase	Verified Route / Target	Scenarios Evaluated	Results
			:--- :---

Detailed Audit Results

1. Authentication Flows

- * **Registration**: Registering the first user automatically inserts them with `role: "admin"` in the SQLite database. Subsequent registrations default to `role: "user"`. Enforces email uniqueness.
- * **Login**: Verifies hashes via `bcrypt`. Successfully signs JWT token containing standard claims (`sub` email, `exp` expiration`). Returns user data payload.
- * **Session Refresher**: Simulating a token countdown in React successfully displays the glassmorphic session warning alert. Pressing "Extend Session" triggers `POST /api/auth/refresh`` on the backend, generating a refreshed token and extending `localStorage` variables.

2. ML Inference Engine & XAI

- * **Single Passenger Prediction**: Sliding parameters and selecting fields computes RF and XGB probabilities. The speedometers reflect the average probability.
- * **SHAP Explanation**: Feature contributions are approximating correctly (e.g. Female Gender yields a high positive influence, Third Pclass yields a high negative influence).
- * **Batch CSV Upload**:
 - * Uploading a correct CSV parses the fields, inputs missing numbers with defaults, and updates the

output preview tables.

* Uploading a dummy file larger than 5MB triggers the new file-size block and returns a `400 Bad Request` toast warning.

3. Reporting & Downloads

* **Analytical PDF**: Clicking the download button queries the `/api/reports/pdf` route using Axios blobs. Generates Helvetica layouts displaying executive model accuracies (81.6% and 83.1%), session rates, and passenger grids.

* **CSV Export**: streams predicting lists inside clean structured comma-separated files.

4. Admin Management

* **User Roles**: Toggling a user's role updates their DB schema, logged instantly in the logs. Admin is prevented from self-demoting.

* **Audit Logs**: Correctly joins User IDs with emails, displaying an audit trails of logins, role updates, and DB backups.

* **JSON System Backup**: Generates complete database dumps in single files.